Power of the Hybrid N-of-1 design under carryover, serial correlation and variance components

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October 29, 2026

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1 Abstract

Background. Aggregated N-of-1 designs can detect a treatment effect with fewer participants than a parallel-group trial. A fair comparison needs designs that estimate the same quantity and analyses whose behavior under the null has been checked.

Methods. We simulated the Hybrid design of Hendrickson et al.[1], an alternating design, a two-period crossover and a parallel-group trial. All designs share a calendar of eight visits over 20 weeks. The drug-response asymptote was set for each design so that all designs estimate the same mean drug response on drug. We ran twelve one-axis sweeps. They cover effect size, carryover half-life, decay form, treatment heterogeneity, within-participant variability,

number of periods, sample size, serial correlation, period length, biomarker correlation and misspecification. Each within-subject dataset was analyzed with a mixed model that built the exposure indicator at two or three assumed half-lives. The parallel group was analyzed by baseline-adjusted ANCOVA. Covariance matrices that were not positive definite were dropped, not repaired. We report bias, empirical and model standard errors, coverage and Monte Carlo standard errors for every cell.

Results. At a mean drug response on drug of -0.50 units and N = 35, power was 0.780 (MCSE 0.013) for the Hybrid and 0.788 (MCSE 0.013) for the parallel group. It was 0.983 (MCSE 0.004) for the alternating design and 0.758 (MCSE 0.014) for the crossover. Pooled over four null configurations, Type I error was 0.054 (MCSE 0.003) for the parallel group and 0.057 (MCSE 0.003) for the Hybrid. It was 0.094 (MCSE 0.003) for the alternating design and 0.148 (MCSE 0.004) for the crossover. Their mean null estimates were -0.06 and -0.14 units. A linear time term did not capture their change in mean from baseline to follow-up. Hybrid power fell from 0.840 to 0.362 as the carryover half-life rose from 0 to 14 days. With treatment heterogeneity of SD 1, parallel-group power fell from 0.810 to 0.412 and Hybrid power from 0.766 to 0.692.

Conclusions. On one estimand at N = 35, the Hybrid had 0.02 less power than a parallel group of the same size, with an MCSE of 0.009 for the difference. Carryover lowered the power of the Hybrid. Treatment heterogeneity lowered the power of the parallel group more. The alternating design and the crossover rejected too often under the null, so their power is not comparable until the analysis model is corrected. The study is exploratory, and its conclusions apply to the parameter values simulated.

Keywords. N-of-1 trials; statistical power; carryover; serial correlation; sensitivity analysis; Monte Carlo simulation; crossover designs.

2 Introduction

Aggregated N-of-1 trials observe each participant on and off treatment and pool the participants in one model[2,3]. Several simulation studies compare them with parallel-group and crossover trials. Blackston et al.[4] simulated three cycles, four variance structures and constant additive carryover. Their parallel-group comparator used one observation per participant. Wang and Schork[5] derived the power of a single participant under AR(1) correlation, period stacking, heteroscedasticity and washout. They had no parallel group. Percha et al.[6] varied the number of blocks, their order, the treatment duration and the sampling frequency under carryover and wash-in. Chen and Chen[7] compared compound-symmetry and AR(1) error structures and paired and mixed-model analyses. Araujo et al.[8] noted that such models omit the participant-by-treatment interaction. Senn[9] set out the variance components that determine the sample size of an N-of-1 series. Gärtner et al.[10] compared analysis methods under carryover. Carrozzo et al.[11] compared a two-arm crossover trial with a meta-analysis of N-of-1 trials. Lin et al.[12] studied bootstrap tests for AB and ABAB series under autocorrelation. Tang and Landes[13] studied t-tests under serial correlation. Shi and Ye[14] studied the power trade-off between crossover and parallel designs under behavioral carryover.

Hendrickson et al.[1] proposed the Hybrid design for validating a predictive biomarker. It combines open-label titration, blinded discontinuation and a brief crossover. We did not find a study of its power for the treatment main effect in a data-generating process (DGP) that includes a placebo-response component, carryover and serial correlation, with the factors varied in turn. This paper reports one.

The paper makes four methodological choices that the 2026-05 version of this study lacked. First, the Hybrid is the published schedule, not a shorter two-path schedule. Second, all designs are put on one estimand, the mean drug response on drug, so that a difference in power is a difference between designs. Third, the analysis half-life is crossed with the data-generating half-life. S2 crosses two assumed half-lives with nine data-generating ones, and S3 and S11 add the data-generating value at 3 days. Fourth, the parallel group is analyzed with a standard arm contrast whose null behavior we check before any comparison.

3 Methods

3.1 Aims

We follow the ADEMP structure of Morris et al.[15]. The study has three aims. The first is to describe the power of the Hybrid design for the treatment main effect as one factor at a time departs from a baseline. The second is to find the regimes in which the Hybrid has more or less power than the comparators on the same estimand. The third is to report Type I error, bias and standard-error calibration for every design. The study is exploratory. No analysis plan was fixed in advance, and the protocol (`00-ademp-protocol.md`) is a working document.

3.2 Designs

All designs have eight follow-up visits on the calendar of the Hybrid, at weeks 4, 8, 9, 10, 11, 12, 16 and 20, except where a sweep changes it. A baseline score is observed before treatment. Table 1 lists the designs. Table 2 gives the drug state of the Hybrid at each visit.

The Hybrid has an open-label phase (weeks 4 and 8, expectancy $\eta_t = 1$), a blinded discontinuation phase (weeks 9 to 12) and a blinded crossover (weeks 16 and 20), with $\eta_t = 0.5$ after week 8. Every path is on drug at week 9. The drug stops after week 9 in two paths and after week 10 in the other two. The crossover is drug first in two paths and placebo first in the other two. The schedule and paths are those of `docs/40`, Section 2, which were checked against the original code at commit 58b32a9.

The comparators are blinded throughout ($\eta_t = 0.5$). The alternating design switches the drug state at every visit and has two paths. The two-period crossover has one assessment at the end of each of two periods, at weeks 10 and 20. The parallel group has two arms, drug at every visit or never. The design `RCT-OL` is the parallel group with $\eta_t = 1$ at every visit. It is the assumption of the 2026-05 version of this study, and it quantifies the effect of the comparator's blinding assumption. The design `Compact2` is the two-path schedule of eight weekly visits used in that version. It is not the published Hybrid and is kept only for S1.

Table 1: Designs. The realized estimand is the mean drug response on drug at the baseline asymptote of 2 units (Section
refestimand).

Design	Paths	Visits	Weeks	Expectancy	Realized estimand
Hybrid	4	8	4, 8, 9, 10, 11, 12, 16, 20	1, 1, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5	-1.51
Compact2	2	8	1, 2, 3, 4, 5, 6, 7, 8	1, 1, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5	-0.84
Alt	2	8	4, 8, 9, 10, 11, 12, 16, 20	0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5	-0.76
XO	2	2	10, 20	0.5, 0.5	-1.84
RCT	2	8	4, 8, 9, 10, 11, 12, 16, 20	0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5	-1.78
RCT-OL	2	8	4, 8, 9, 10, 11, 12, 16, 20	1, 1, 1, 1, 1, 1, 1, 1	-1.78

Table 2: Drug state of the Hybrid by path and week. Expectancy is 1 at weeks 4 and 8 and 0.5 afterward.

path	4	8	9	10	11	12	16	20
1	on	on	on	on	off	off	on	off
2	on	on	on	on	off	off	off	on
3	on	on	on	off	off	off	on	off
4	on	on	on	off	off	off	off	on

The sample size N is the total across paths. With $N = 35$ and four paths, the paths receive 9, 9, 9 and 8 participants. The alternating design receives 18 and 17. The parallel group receives 18 and 17. A parallel group of 17 participants per arm is small, and all conclusions refer to that size unless a sweep changes N.

3.3 Data-generating process

The outcome is $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$. The baseline score BL_i has mean 8 and SD 1.5. The three components are the natural-history component TV, the placebo-response component PB and the drug-response component BR. They are reductions in symptoms, so a drug effect is negative. Each component has a modified Gompertz mean

$$G(t; m, d, r) = m \, \frac{\exp(-d\, e^{-rt}) - e^{-d}}{1 - e^{-d}},$$

with asymptote m, displacement $d = 1$ and rate $r = 0.3$ per week. The asymptotes are $m = 0.5$ for TV, 1 for PB and a design-specific value for BR (next section). The component SDs are 0.5, 0.5 and 1. The TV mean runs on calendar weeks, the PB mean on the time under positive expectancy scaled by η_t, and the BR mean on the time on drug in the current run. The PB SD is also scaled by η_t. A participant who is re-exposed starts the BR curve again from zero. Within each component, the correlation between visits s and t is $\rho^{|w_s - w_t|}$ with $\rho = 0.3$ per week. The correlation between components is 0.1 at the same visit and $0.05\,\rho^{|w_s - w_t|}$ at different visits. The biomarker B_i has mean 120 and SD 15. Its correlation with BR at on-drug visits is $c_{bm} = 0.3$ (the correlation encoding). The mean-shift encoding

shifts BR at on-drug visits by $c_{bm}\, b_i\, \sigma_{BR}$, with b_i the standardized biomarker. S10 compares the two encodings.

Carryover acts on the BR mean. At an off-drug visit after exposure, the mean is the previous visit's mean multiplied by $\phi(t_{sd})$. The decay family ϕ is exponential, $e^{-\lambda t}$ with $\lambda = \ln 2/t_{1/2}$, linear, Weibull or power (Section 3.8). The baseline half-life is 3 days. This rule adds the decayed mean of the previous visit, which already contains earlier carryover. Across consecutive off-drug visits it compounds, so the realized decay is faster than the nominal half-life implies (`docs/53`). The half-life axes are therefore nominal.

Treatment heterogeneity enters through τ. Each participant draws $v_i \sim \mathcal{N}(0, \tau^2)$ once and the drug response at visit t is shifted by $v_i \Psi_{it}$, with Ψ_{it} the mean drug response divided by its asymptote. The mean effect is unchanged. The baseline has $\tau = 0$.

Units. We quote the carryover half-life in days, because the 3-day baseline matches the pharmacokinetic scale of the prazosin application. The companion papers on the biomarker interaction quote it in weeks on the grid $\{0, 0.5, 1\}$. One week is seven days, so that grid is $\{0, 3.5, 7\}$ days here. The 3-day baseline is $3/7 \approx 0.43$ week. Visits are weekly in both conventions. Whether paper 05 should move to weeks is an open decision (Section 5.4).

3.4 Estimand

The designs expose participants to the drug for different times, and the Gompertz response restarts at each re-exposure. A fixed asymptote m therefore gives a different mean effect in each design. The 2026-05 version compared designs at one asymptote. We define the realized estimand

$$\Delta_{\text{on}} = -\frac{1}{n_{\text{on}}} \sum_{q=1}^{P} \sum_{t:\, D_{qt}=1} \mu_{BR,qt},$$

the mean of the drug-response mean $\mu_{BR,qt}$ over the n_{on} on-drug follow-up visits of the P paths. It depends on the design and on m only, because the carryover rule acts only at off-drug visits. It is linear in m. In every sweep except S1 the asymptote m is set for each design so that Δ_{on} equals -0.5 outcome units. In S1, Δ_{on} is the swept axis. The stored value of Δ_{on} is in column `delta_on` of every artifact.

The coefficient β_D that an analysis estimates need not equal Δ_{on}. For the parallel group with the ANCOVA contrast it does. For the within-subject designs the model includes linear time and, in the Hybrid, expectancy, so β_D differs from Δ_{on} by an amount that depends on the design. We take as the target of bias and coverage the large-sample value of β_D for each analysis, the pooled fit to all simulated participants of a cell (column `ETbeta`). We report both the bias against this value and the difference from Δ_{on}.

3.5 Analysis models

The within-subject designs are fitted with `nlme::lme`[16,17]. The model has a random intercept for the participant and a continuous-time AR(1) residual correlation, `corCAR1(form = ~t |`

Table 3: Models fitted. The table lists the formulas stored in the S1 artifact when it is available and otherwise the formulas the code builds.

Design	Model fitted
Hybrid	Sx ~ bm + t + Dbc + De [random = ~1|ptID; corCAR1(~t|ptID)]
Compact2	Sx ~ bm + t + Dbc + De [random = ~1|ptID; corCAR1(~t|ptID)]
Alt	Sx ~ bm + t + Dbc [random = ~1|ptID; corCAR1(~t|ptID)]
XO	Sx ~ bm + t + Dbc [random = ~1|ptID; corCAR1(~t|ptID)]
RCT	y_bar ~ arm + BL [OLS; y_bar = follow-up mean]
RCT-OL	y_bar ~ arm + BL [OLS; y_bar = follow-up mean]

`ptID)`. The fixed part is

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_\eta \eta_t + u_i + \varepsilon_{it}.$$

The term in η_t enters only when the design has more than one value of expectancy, which is the Hybrid. The baseline row ($w = 0$, $D = 0$, $\eta = 0$) is included. The engine counts expectancy values over the follow-up visits only. In the blinded designs the baseline row has $\eta = 0$ and every follow-up visit has $\eta = 0.5$, so their model has no term for this change (Section 4.14). The exposure indicator is $D_{bc,it} = 1$ on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after exposure and 0 before it. The assumed half-life $t^a_{1/2}$ is a property of the analysis. It is 0 (binary indicator), 0.2 week (the project policy value when the half-life is fixed), or 3/7 week (the data-generating half-life). Which values a sweep uses is listed with the sweep. The target is β_D. The stored column `fit_formula` records the model actually fitted in every replicate. Table 3 lists the formulas.

The parallel group is analyzed by ANCOVA. Each participant's follow-up observations are averaged and the mean is regressed on the arm indicator and the baseline score by ordinary least squares. The arm coefficient is β_D. The 2026-05 version fitted the mixed model above, with a baseline row in which $D = 0$ for both arms. That model has one shared linear time term. The engine keeps it as `parallel_analysis = 'legacy'` only to reproduce stored runs. Section 4.1 reports the check of both analyses.

3.6 Performance measures

For each cell we report, with Monte Carlo standard errors (MCSE) after Morris et al.[15], the rejection rate at $\alpha = 0.05$ (power, or Type I error when the effect is zero), the bias $\overline{\hat{\beta}_D} - \beta_D^\infty$ with β_D^∞ the large-sample value, the empirical SD of $\hat{\beta}_D$, the mean model SE, the ratio κ of the two, the coverage of the 95% interval around β_D^∞ and the share of fits that failed. A Type I error is called nominal when it lies within two MCSE of 0.05, with the MCSE computed at 0.05. The band is 0.05 ± 0.019 at 500 replicates, ± 0.014 at 1,000 and ± 0.010 at 2,000. The MCSE of a power near 0.5 is 0.022 at 500 replicates and 0.011 at 2,000. Bias, SEs, κ and coverage for every cell are in `analysis/data/derived_data/sensitivity/cell-summaries.csv`, written by `16-summarize-all.R`.

Table 4: S8. Smallest eigenvalue of the correlation matrix by AR(1) correlation (rows) and design (columns). A negative value is infeasible.

rho	Hybrid	Alt	RCT
0.0	0.3136	0.3801	0.1410
0.3	0.3953	0.3218	0.2718
0.5	0.2803	0.1917	0.2909
0.7	0.1273	0.0566	0.1332
0.9	-0.0020	-0.0676	0.0049

Table 5: S10. Smallest eigenvalue by biomarker correlation (rows) and design with half-life in days (columns), correlation encoding. A negative value is infeasible.

c_bm	Hybrid, 0 d	Alt, 0 d	RCT, 0 d	Hybrid, 3 d	Alt, 3 d	RCT, 3 d
0.0	0.4915	0.4915	0.4915	0.4915	0.4915	0.4915
0.2	0.4778	0.4082	0.4886	0.4839	0.4312	0.4886
0.4	0.1819	0.1242	0.0043	0.1948	0.1604	0.0043
0.6	-0.2453	-0.2400	-0.5441	-0.2351	-0.2087	-0.5441

The large-sample value comes from one fit to all simulated participants of a cell. It differed from the mean of the replicate estimates by more than four MCSE in 21 of 405 cells. All of them are alternating designs. The largest difference was 0.166 units. Bias and coverage against `ETbeta` are not interpretable in those cells. We therefore also report the mean estimate minus Δ_{on}, which at the null is the mean estimate itself.

3.7 Feasibility of the covariance matrix

A cell is feasible when its covariance matrix is positive definite. The engine checks this for every path of every cell before the cell runs. It does not repair a matrix that fails the check. It drops the cell, and the artifact records the cell, the smallest eigenvalue of the correlation matrix and the reason. The column `pd_repaired` is stored in every replicate row and is `FALSE` throughout. The 2026-05 version repaired such matrices silently. In S8 and S10, the cells that fail are therefore absent. Script `17-feasibility.R` computes the smallest eigenvalue for these two grids without simulation.

In S8, 2 of 15 design and correlation cells are infeasible: Hybrid at $\rho = 0.9$ (smallest eigenvalue -0.002) and Alt at $\rho = 0.9$ (smallest eigenvalue -0.068). In S10 under the correlation encoding, 6 of 24 cells are infeasible: $c_{bm} = 0.6$ for Hybrid, Alt, and RCT. The smallest eigenvalue of the feasible cells is 0.004. The same check runs inside every sweep, and the artifacts list the dropped cells.

3.8 The twelve sweeps

Each sweep varies one factor (S4 and S11 vary two blocks) from the baseline of Table 6. Table 7 lists the grids.

Table 6: Baseline of the sweeps.

Quantity	Baseline
N (total)	35
Visits	8, weeks 4 to 20
Mean drug response on drug	-0.5
Carryover half-life	3 days
Decay form	exponential
AR(1) correlation per week	0.3
Heterogeneity SD	0
Biomarker correlation	0.3 (correlation encoding)
Assumed half-life	0.2 week

Table 7: Sweeps. The number of cells is the planned number before infeasible cells are dropped.

Sweep	Axis	Grid	Designs	Replicates	Cells
S1	Mean drug response on drug	0, -0.25, -0.5, -0.75, -1, -1.5, -2	Hybrid, Compact2, Alt, XO, RCT, RCT-OL	1,000	42
S2	Carryover half-life, by assumed half-life	0, 0.5, 1, 2, 3, 5, 7, 10, 14 days; assumed 0, 0.2 week	Hybrid, Alt, XO, RCT	500	36
S3	Decay form, by assumed half-life	exponential, linear, Weibull k = 0.8 and 1.5, power; assumed 0, 0.2, 3/7 week	Hybrid, Alt, RCT	500	15
S4	Heterogeneity SD τ; baseline SD	τ: 0, 0.25, 0.5, 0.75, 1; baseline SD: 0.5, 1, 1.5, 2, 3	Hybrid, Alt, RCT	500	30
S5	Common multiplier on the three component SDs	0.5, 0.75, 1, 1.25, 1.5	Hybrid, Alt, RCT	500	15
S6	Periods in 16 weekly visits	2, 4, 8, 16 periods; parallel group	Alt with 2 to 16 periods, RCT	500	5
S7	Total N	16, 24, 35, 60, 80, 120	Hybrid, Alt, RCT	500	18
S8	AR(1) correlation per week	0, 0.3, 0.5, 0.7, 0.9	Hybrid, Alt, RCT	500	15
S9	Period length by half-life	3, 5, 7, 10, 14 days by half-life 0.5, 3, 7 days	Alt	500	15
S10	Biomarker correlation by encoding and half-life	c_{bm} 0, 0.2, 0.4, 0.6; half-life 0, 3 days	Hybrid, Alt, RCT	500	48
S11	DGP carryover by DGP AR(1), by assumed half-life	off or 3 days by off or $\rho = 0.7$; null and baseline effect	Hybrid, Alt, RCT	500	24
S12	Null effect, four configurations	baseline SD 1.5, 1.5, 0.75, 2; drug-response SD 1, 1, 1.5, 0.5; half-life 0, 3, 3, 3 days	Hybrid, Alt, XO, RCT	2,000	16

The precedents are as follows. S1 follows Blackston et al.[4]. S2 and S9 follow Blackston et al., Wang and Schork[5] and Percha et al.[6]. S3 extends the carryover forms of the engine, and Gärtner et al.[10] compared analysis methods under carryover but not decay forms. S4 and S5 follow the variance components of Senn[9] and Zucker et al.[18]. S6 follows Percha et al. S8 follows Wang and Schork and Lin et al.[12]. S12 follows Morris et al.

S3 uses the following decay families. Linear is $\max(0, 1 - t/(2t_{1/2}))$. Weibull is $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$. Power is $\max(0, 1 - t/(3t_{1/2}))^p$ with $p = 1.5$. The multiplier equals 0.5 at $t = t_{1/2}$ for the exponential, linear and Weibull forms. For the power form it equals $(2/3)^{1.5} = 0.544$.

S4 has two blocks. The first varies τ at the baseline SD of the baseline score. The second varies that SD at $\tau = 0$. The baseline SD is a random intercept that enters every visit equally, so it is not treatment heterogeneity. S5 multiplies the SDs of all three components, so the within-participant SD of the outcome scales with the multiplier. S6 fixes the trial at 16 weekly visits and varies the number of equal periods. The number of switches is the number of periods minus one. The BR response restarts in each period, and the asymptote is set so that Δ_{on} is the same in every design.

The sample-size sweep S7 keeps the allocation rule of the engine. The null calibration S12 sets the drug-response asymptote to zero and keeps $c_{bm} = 0.3$. S12b refits the null data of the crossover and the alternating design without the serial-correlation structure. S14 is the check of the parallel-group analyses (Section 4.1).

3.9 Software and seeds

The simulation engine is the R package `nof1power` (version 0.1.0), which is part of this project. It is not an external package of Hendrickson et al. The drivers are in `analysis/scripts/nof1-design-sensitivity/`. Each driver calls `set.seed(42 + id)` once, where `id` is the sweep number. The cells then run in parallel with `furrr` and `seed = TRUE`, which gives each cell its own random-number stream derived from the master seed. Results from the container and from a local session are not draw-for-draw identical, because the linear-algebra libraries differ. Each artifact stores the engine version, the repository revision, the seed, the replicate count and the date. The sweeps were run in one job on the simulation host, from commit 6485001 with `nof1power` 0.1.0, in 2 h 45 min. The revision field of these artifacts is empty, because the host container had no git metadata. The check of Section 4.14 was run locally from the working tree.

4 Results

All numbers in this section are generated from the stored artifacts in `analysis/data/derived_data/sensitivit`
The document reads an artifact only if it carries the run metadata of the corrected drivers. Every sweep artifact used here does. No sweep had a failed fit, and no covariance matrix was repaired. The baseline cell recurs in several sweeps with independent draws, so differences between sweeps include Monte Carlo error. Comparisons are made within a sweep.

Table 8: Parallel-group analyses. The mean estimate at the null should be zero. The bias is the mean estimate minus the realized estimand. kappa is the mean model SE divided by the empirical SD.

Design	Analysis	Effect	rho	Mean estimate	Bias vs estimand	Bias/MCSE	Empirical SE	Model SE	kappa	Rejection rate
RCT	ancova	-0.50	0.0	-0.500	0.000	0.0	0.150	0.147	0.980	0.906
RCT	ancova	-0.50	0.3	-0.504	-0.004	-0.8	0.176	0.172	0.972	0.800
RCT	ancova	-0.50	0.7	-0.494	0.006	0.8	0.248	0.238	0.956	0.525
RCT	ancova	0.00	0.0	0.008	0.008	1.7	0.150	0.147	0.977	0.050
RCT	ancova	0.00	0.3	0.001	0.001	0.1	0.173	0.171	0.990	0.050
RCT	ancova	0.00	0.7	-0.002	-0.002	-0.2	0.237	0.238	1.003	0.049
RCT	legacy	-0.50	0.0	-0.827	-0.327	-52.2	0.198	0.285	1.438	0.902
RCT	legacy	-0.50	0.3	-0.830	-0.330	-48.6	0.215	0.288	1.340	0.895
RCT	legacy	-0.50	0.7	-0.779	-0.279	-36.8	0.240	0.284	1.184	0.817
RCT	legacy	0.00	0.0	-0.367	-0.367	-56.6	0.205	0.285	1.393	0.169
RCT	legacy	0.00	0.3	-0.369	-0.369	-55.0	0.212	0.288	1.357	0.176
RCT	legacy	0.00	0.7	-0.358	-0.358	-48.5	0.234	0.284	1.217	0.196
RCT-OL	ancova	-0.50	0.0	-0.498	0.002	0.3	0.167	0.159	0.949	0.840
RCT-OL	ancova	-0.50	0.3	-0.503	-0.003	-0.4	0.187	0.186	0.992	0.746
RCT-OL	ancova	-0.50	0.7	-0.504	-0.004	-0.5	0.267	0.258	0.965	0.488
RCT-OL	ancova	0.00	0.0	-0.006	-0.006	-1.2	0.164	0.160	0.974	0.065
RCT-OL	ancova	0.00	0.3	0.002	0.002	0.3	0.182	0.186	1.020	0.046
RCT-OL	ancova	0.00	0.7	0.002	0.002	0.2	0.264	0.257	0.976	0.048
RCT-OL	legacy	-0.50	0.0	-0.998	-0.498	-70.8	0.222	0.306	1.377	0.967
RCT-OL	legacy	-0.50	0.3	-0.994	-0.494	-66.7	0.234	0.309	1.321	0.947
RCT-OL	legacy	-0.50	0.7	-0.951	-0.451	-54.0	0.264	0.307	1.160	0.905
RCT-OL	legacy	0.00	0.0	-0.544	-0.544	-76.0	0.226	0.306	1.354	0.404
RCT-OL	legacy	0.00	0.3	-0.531	-0.531	-73.1	0.230	0.309	1.346	0.373
RCT-OL	legacy	0.00	0.7	-0.513	-0.513	-62.4	0.260	0.308	1.185	0.362

4.1 Parallel-group analyses under the null

We checked the two analyses of the parallel group before comparing designs. The check used the blinded and the open-label parallel group, three AR(1) correlations and two effects (the null and the baseline effect).

Under the null, the mean estimate of the mixed-model analysis of the parallel group ranged from -0.544 to -0.358, and its Type I error from 0.169 to 0.404. The largest |mean|/MCSE was 76.0 and κ ranged from 1.185 to 1.393. The ANCOVA arm contrast had mean estimates from -0.006 to 0.008 (largest |mean|/MCSE 1.7), κ from 0.974 to 1.020, and Type I error from 0.046 to 0.065. Each cell has 1,000 replicates.

4.2 S1. Effect size

We varied the mean drug response on drug over 7 values from 0 to -2.00 outcome units, with 1,000 replicates per cell. The asymptote of the drug-response curve was set for each design so that every design had the same value. At the null, Type I error was Alt 0.090 (MCSE 0.009), Compact2 0.053 (MCSE 0.007), Hybrid 0.078 (MCSE 0.008), RCT 0.048 (MCSE 0.007), RCT-OL 0.062 (MCSE 0.008), and XO 0.137 (MCSE 0.011). Of 6 designs, 3 were above and 0 below the band of two MCSE around 0.05. At the null the estimand is zero. The mean estimate was Alt -0.068 (MCSE 0.004), Compact2 -0.019 (MCSE 0.005), Hybrid 0.019 (MCSE 0.006), RCT 0.009 (MCSE 0.005), RCT-OL -0.008 (MCSE 0.006), and XO -0.128 (MCSE 0.009). The ratio of model to empirical SE (κ) at the null was Alt 0.975, Compact2

Table 9: S1. Rejection rate by mean drug response on drug (rows) and design (columns). Monte Carlo standard error at most 0.016.

Effect	Alt	Compact2	Hybrid	RCT	RCT-OL	XO
-2.00	1.000	1.000	1.000	1.000	1.000	1.000
-1.50	1.000	1.000	1.000	1.000	1.000	1.000
-1.00	1.000	1.000	1.000	1.000	0.999	0.991
-0.75	1.000	0.999	0.981	0.990	0.963	0.934
-0.50	0.983	0.892	0.780	0.788	0.744	0.758
-0.25	0.619	0.405	0.273	0.276	0.264	0.413
0.00	0.090	0.053	0.078	0.048	0.062	0.137

Table 10: S1. Calibration at the null and at the baseline effect. Mean - estimand is the mean estimate minus the realized estimand. Bias and coverage are against the large-sample value ‘ETbeta‘, which is unreliable for the alternating design (Section refperf).

Design	Effect	Mean - estimand	Bias/MCSE	Bias	Empirical SE	Model SE	kappa	Coverage	Failed
Alt	-0.5	0.005	16.9	0.067	0.126	0.121	0.960	0.899	0.000
Alt	0.0	-0.068	0.1	0.000	0.122	0.119	0.975	0.939	0.000
Compact2	-0.5	-0.079	0.0	0.000	0.181	0.175	0.962	0.940	0.000
Compact2	0.0	-0.019	2.2	0.012	0.173	0.172	0.992	0.948	0.000
Hybrid	-0.5	0.044	-0.2	-0.001	0.168	0.168	1.002	0.953	0.000
Hybrid	0.0	0.019	-0.0	-0.000	0.178	0.166	0.934	0.924	0.000
RCT	-0.5	0.007	-0.2	-0.001	0.173	0.173	1.000	0.953	0.000
RCT	0.0	0.009	-0.0	-0.000	0.173	0.172	0.995	0.956	0.000
RCT-OL	-0.5	-0.005	-0.2	-0.001	0.193	0.186	0.961	0.945	0.000
RCT-OL	0.0	-0.008	-0.3	-0.002	0.192	0.185	0.964	0.939	0.000
XO	-0.5	-0.147	0.0	0.000	0.273	0.230	0.843	0.895	0.000
XO	0.0	-0.128	0.0	0.000	0.277	0.232	0.839	0.906	0.000

0.992, Hybrid 0.934, RCT 0.995, RCT-OL 0.964, and XO 0.839. The smallest grid effect with power of at least 0.80 was Alt -0.50, Compact2 -0.50, Hybrid -0.75, RCT -0.75, RCT-OL -0.75, and XO -0.75. At the baseline effect of -0.50, power was Alt 0.983 (MCSE 0.004), Compact2 0.892 (MCSE 0.010), Hybrid 0.780 (MCSE 0.013), RCT 0.788 (MCSE 0.013), RCT-OL 0.744 (MCSE 0.014), and XO 0.758 (MCSE 0.014).

4.3 S2. Carryover half-life

We varied the data-generating half-life over 9 values from 0 to 14 days. Each dataset was analyzed with the binary on/off indicator (analysis half-life 0) and with the decaying indicator at 0.2 week. With the analysis half-life at 0.00 week, power at 0 and 14 days was Alt 0.990 and 0.792, Hybrid 0.842 and 0.346, RCT 0.802 and 0.796, and XO 0.748 and 0.730. With the analysis half-life at 0.20 week, power at 0 and 14 days was Alt 0.990 and 0.796, Hybrid 0.840 and 0.362, RCT 0.802 and 0.796, and XO 0.748 and 0.730. The two analyses differed in power by at most 0.026 in any cell. At analysis half-life 0.2 week, the mean estimate moved from 0 to 14 days as follows: Alt from -0.537 to -0.343, Hybrid from -0.503 to -0.267, RCT from

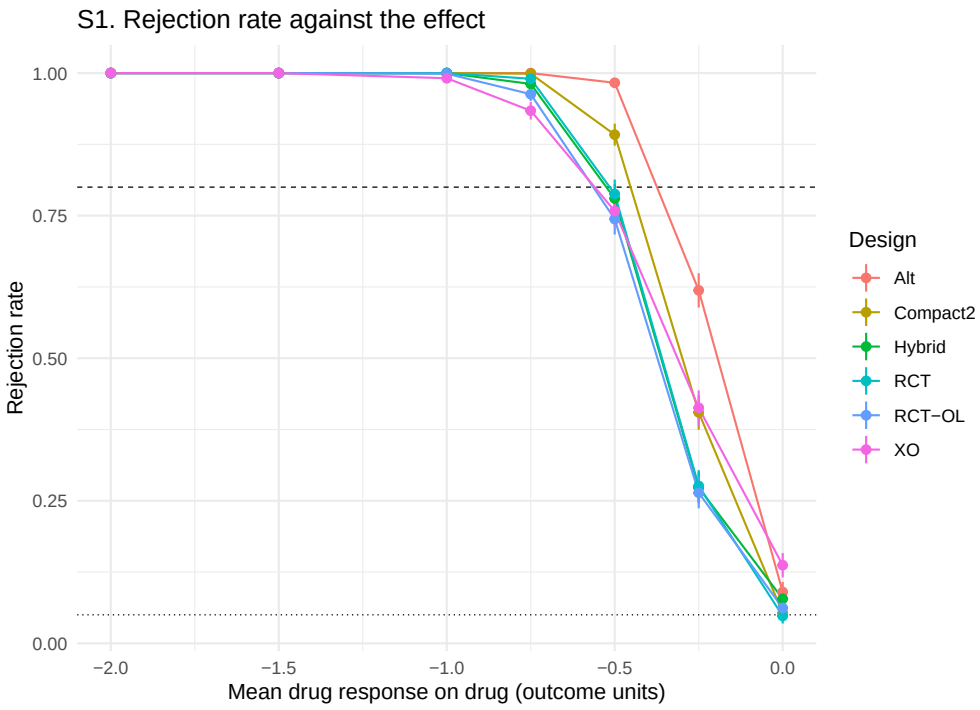


Figure 1: S1. Rejection rate against the mean drug response on drug, with 95 percent Monte Carlo intervals.

-0.499 to -0.499, and XO from -0.648 to -0.625. The estimand was -0.50 throughout. Each cell has 500 replicates. The MCSE of a power near 0.5 is 0.022.

4.4 S3. Decay form

We varied the form of the data-generating carryover over 5 forms at a half-life of 3 days. The analysis always built the decaying indicator from an exponential, at three half-lives. With analysis half-life 0.00 week, power ranged across decay forms from 0.738 to 0.988 (Alt 0.970 to 0.988, Hybrid 0.738 to 0.820, and RCT 0.798 to 0.818). With analysis half-life 0.20 week, power ranged across decay forms from 0.740 to 0.988 (Alt 0.974 to 0.988, Hybrid 0.740 to 0.824, and RCT 0.798 to 0.818). With analysis half-life 0.43 week, power ranged across decay

Table 11: S2. Power by data-generating half-life (rows) and design with assumed half-life (columns). Monte Carlo standard error at most 0.022.

Half-life (days)	Alt, 0.0 wk	Alt, 0.2 wk	Hybrid, 0.0 wk	Hybrid, 0.2 wk	RCT, 0.0 wk	RCT, 0.2 wk	XO, 0.0 wk	XO, 0.2 wk
0.0	0.990	0.990	0.842	0.840	0.802	0.802	0.748	0.748
0.5	0.992	0.992	0.838	0.840	0.826	0.826	0.696	0.696
1.0	0.994	0.994	0.810	0.816	0.792	0.792	0.758	0.758
2.0	0.988	0.988	0.836	0.834	0.826	0.826	0.710	0.710
3.0	0.974	0.976	0.750	0.752	0.776	0.776	0.732	0.732
5.0	0.938	0.944	0.690	0.698	0.766	0.766	0.732	0.732
7.0	0.926	0.930	0.610	0.620	0.822	0.822	0.734	0.734
10.0	0.846	0.860	0.474	0.500	0.812	0.812	0.764	0.764
14.0	0.792	0.796	0.346	0.362	0.796	0.796	0.730	0.730

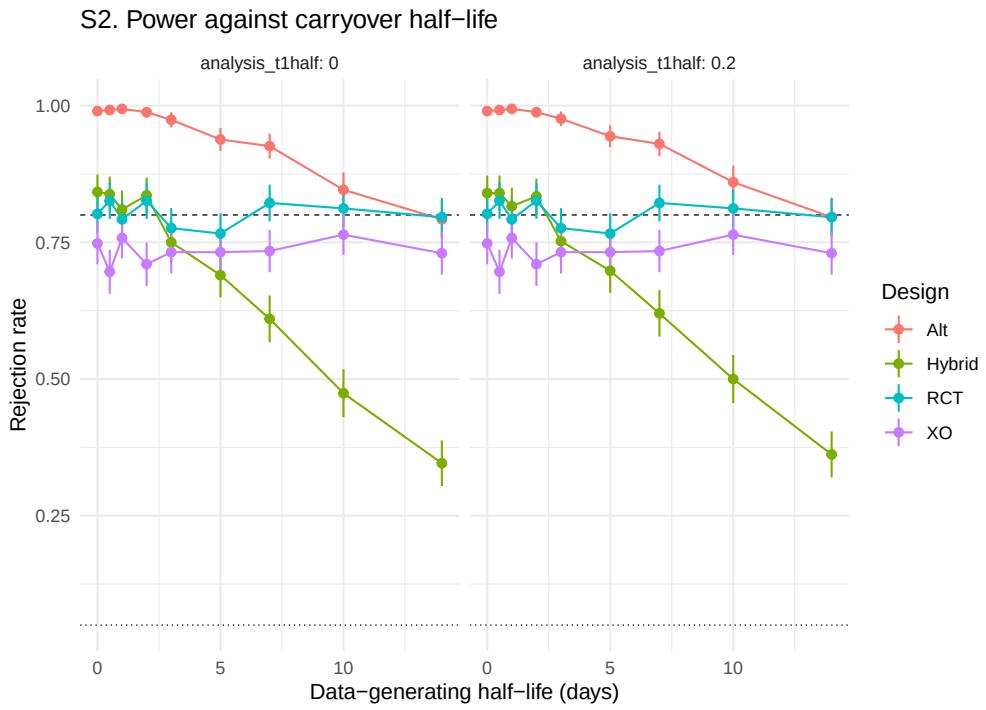


Figure 2: S2. Power against the data-generating half-life, by assumed half-life.

Table 12: S3. Power by decay form (rows) and design with assumed half-life (columns). Monte Carlo standard error at most 0.020.

Decay form	Alt, 0.00 wk	Alt, 0.20 wk	Alt, 0.43 wk	Hybrid, 0.00 wk	Hybrid, 0.20 wk	Hybrid, 0.43 wk	RCT, 0.00 wk	RCT, 0.2
exponential	0.974	0.978	0.992	0.820	0.824	0.844	0.804	0.804
linear	0.988	0.988	0.994	0.814	0.816	0.824	0.800	0.800
power	0.978	0.978	0.984	0.788	0.798	0.802	0.818	0.818
weibull_k=0.8	0.970	0.974	0.984	0.738	0.740	0.760	0.806	0.806
weibull_k=1.5	0.986	0.988	0.990	0.788	0.788	0.800	0.798	0.798

forms from 0.760 to 0.994 (Alt 0.984 to 0.994, Hybrid 0.760 to 0.844, and RCT 0.798 to 0.818). Each cell has 500 replicates.

4.5 S4. Treatment heterogeneity and baseline SD

We varied the SD of participant-level heterogeneity of the drug effect (τ) over 5 values and, separately, the SD of the baseline score over 5 values. Across τ, power was Alt 0.990 at 0 and 0.970 at 1, Hybrid 0.766 at 0 and 0.692 at 1, and RCT 0.810 at 0 and 0.412 at 1. Over the same range the SEs were: Alt empirical SE 0.123 to 0.131, model SE 0.121 to 0.124, κ 0.945 and coverage 0.938 at $\tau = 1$, Hybrid empirical SE 0.171 to 0.216, model SE 0.167 to 0.176, κ 0.815 and coverage 0.888 at $\tau = 1$, and RCT empirical SE 0.172 to 0.284, model SE 0.173 to 0.280, κ 0.986 and coverage 0.946 at $\tau = 1$. The engine scales the heterogeneity by the drug-response curve divided by its asymptote. At the matched estimand the asymptote differs by design, so the heterogeneity SD at the mean on-drug visit is τ times 0.76 for Hybrid, 0.38 for Alt, and 0.89 for RCT. Across the baseline SD, power was Alt 0.986 at 0.5 and 0.988

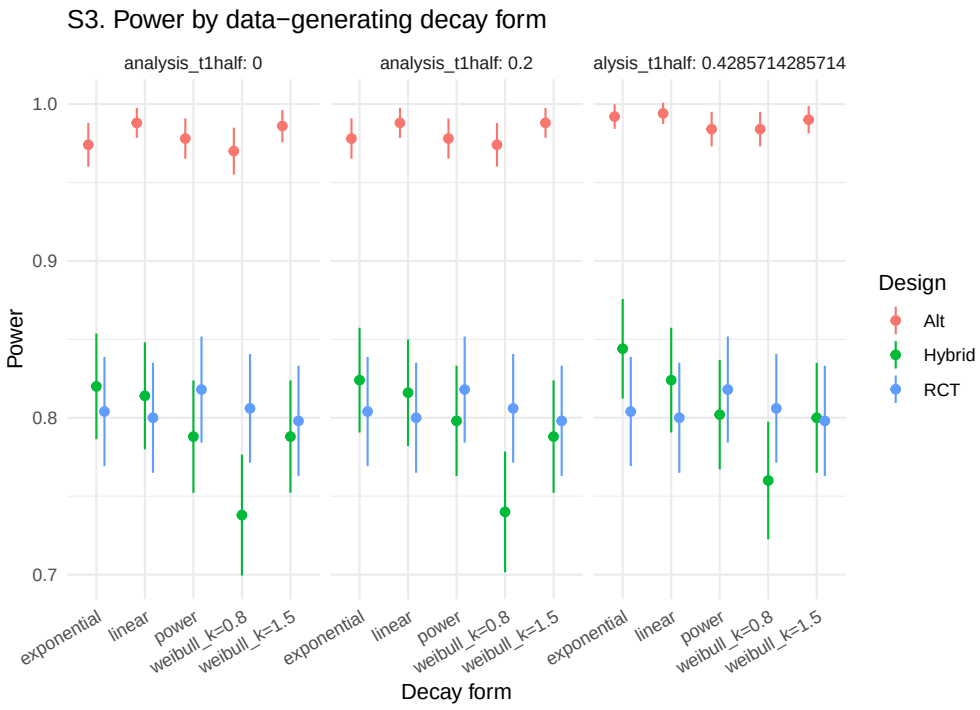


Figure 3: S3. Power by data-generating decay form, by assumed half-life.

Table 13: S4. Power by heterogeneity SD (rows) and design (columns). Monte Carlo standard error at most 0.022.

tau	Alt	Hybrid	RCT
0.00	0.990	0.766	0.810
0.25	0.986	0.738	0.764
0.50	0.986	0.764	0.622
0.75	0.978	0.722	0.514
1.00	0.970	0.692	0.412

at 3, Hybrid 0.758 at 0.5 and 0.748 at 3, and RCT 0.798 at 0.5 and 0.810 at 3. Each cell has 500 replicates.

4.6 S5. Within-participant variability

We multiplied the SDs of all three components by a common factor from 0.5 to 1.5 (5 values). Power at the smallest and largest factor was Alt 1.000 and 0.750, Hybrid 1.000 and 0.476, and RCT 1.000 and 0.464. Each cell has 500 replicates.

4.7 S6. Number of periods at fixed trial length

We varied the number of equal periods of the alternating design over 2, 4, 8, 16, with 16 weekly visits throughout. Power of the alternating design was 0.988 with 2 periods, 0.998

Table 14: S4. Power by baseline SD (rows) and design (columns). Monte Carlo standard error at most 0.022.

Baseline SD	Alt	Hybrid	RCT
0.5	0.986	0.758	0.798
1.0	0.980	0.778	0.778
1.5	0.986	0.780	0.796
2.0	0.978	0.782	0.764
3.0	0.988	0.748	0.810

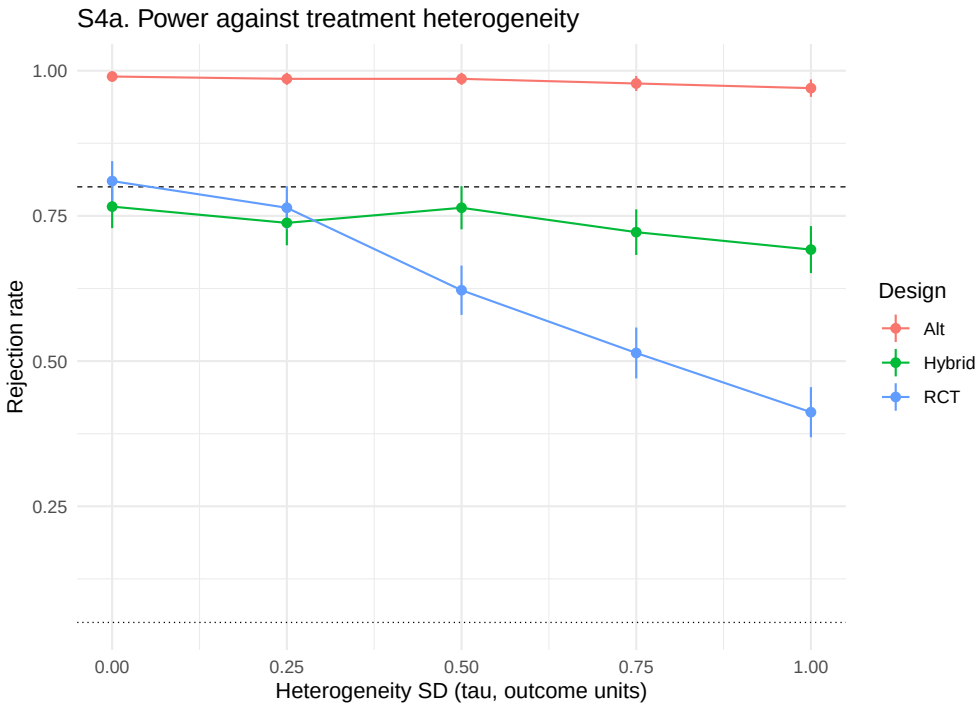


Figure 4: S4. Power against the heterogeneity SD.

Table 15: S5. Power by SD multiplier (rows) and design (columns). Monte Carlo standard error at most 0.022.

Multiplier	Alt	Hybrid	RCT
0.50	1.000	1.000	1.000
0.75	1.000	0.948	0.960
1.00	0.982	0.802	0.794
1.25	0.898	0.612	0.592
1.50	0.750	0.476	0.464

Table 16: S6. Power by design and number of periods.

Design	Periods	Power (MCSE)	kappa
Alt-P16	16	1.000 (MCSE 0.000)	1.004
Alt-P2	2	0.988 (MCSE 0.005)	0.968
Alt-P4	4	0.998 (MCSE 0.002)	1.015
Alt-P8	8	1.000 (MCSE 0.000)	1.000
RCT	1	0.934 (MCSE 0.011)	0.983

Table 17: S7. Power by total N (rows) and design (columns). Monte Carlo standard error at most 0.022.

N	Alt	Hybrid	RCT
16	0.802	0.454	0.470
24	0.902	0.634	0.630
35	0.976	0.758	0.834
60	1.000	0.946	0.964
80	1.000	0.984	0.992
120	1.000	1.000	0.998

with 4 periods, 1.000 with 8 periods, and 1.000 with 16 periods. Power of the parallel group with the same 16 visits was 0.934 (MCSE 0.011). The realized estimand was -0.500 for all designs. Each cell has 500 replicates.

4.8 S7. Sample size

We varied the total sample size over 16, 24, 35, 60, 80, 120. The smallest N in the grid with power of at least 0.80 was Alt 16, Hybrid 60, and RCT 35. Each cell has 500 replicates.

4.9 S8. Serial correlation

We varied the AR(1) correlation per week over 0, 0.3, 0.5, 0.7, 0.9. 2 of 15 cells had a covariance matrix that was not positive definite and were dropped, not repaired (Hybrid at 0.9, smallest eigenvalue -0.002 and Alt at 0.9, smallest eigenvalue -0.068). Power at the smallest and largest run correlation was Alt 0.982 at 0 and 0.996 at 0.7, Hybrid 0.764 at 0 and 0.910 at 0.7, and RCT 0.922 at 0 and 0.336 at 0.9. Each cell has 500 replicates.

4.10 S9. Period length and half-life

We crossed 5 period lengths (3, 5, 7, 10, 14 days) with 3 half-lives (0.5, 3, 7 days). The stored half-life takes 3 distinct values in 15 cells. Power ranged from 0.666 to 1.000. At the longest period, power at the shortest and longest half-life was 0.988 and 0.962. Each cell has 500 replicates.

Table 18: S8. Power by AR(1) correlation (rows) and design (columns). Dropped cells are absent. Monte Carlo standard error at most 0.022.

AR(1) correlation	Alt	Hybrid	RCT
0.0	0.982	0.764	0.922
0.3	0.966	0.756	0.794
0.5	0.990	0.782	0.714
0.7	0.996	0.910	0.526
0.9			0.336

Table 19: S9. Power by period length in days (rows) and half-life in days (columns). Monte Carlo standard error at most 0.021.

Period (days)	0.5	3	7
3	1.000	0.960	0.666
5	1.000	0.966	0.748
7	1.000	0.966	0.802
10	0.992	0.988	0.900
14	0.988	0.980	0.962

4.11 S10. Biomarker correlation and encoding

The sweep stored the main-effect coefficient. We varied the biomarker-response correlation c_{bm} over 0, 0.2, 0.4 and 0.6, the carryover half-life over 0 and 3 days, and the encoding of the interaction (the correlation encoding `mvn` and the mean-shift encoding `mean_moderation`). 6 of 48 cells were not positive definite and were dropped: $c_{bm} = 0.6$ under the correlation encoding, in every design. Across the cells that ran, power ranged from 0.592 to 0.992 (Alt 0.912 to 0.992, Hybrid 0.704 to 0.832, and RCT 0.592 to 0.824). Each cell has 500 replicates.

4.12 S11. Misspecification factorial

We crossed data-generating carryover (off, 3 days) with data-generating AR(1) (off, 0.7), at the null and at the baseline effect, and analyzed each dataset with three analysis half-lives (0, 0.2 week and the data-generating 3/7 week). Type I error ranged over cells and analyses from Alt 0.058 to 0.150 (7 of 12 cells outside the band), Hybrid 0.038 to 0.080 (3 of 12 cells outside the band), and RCT 0.038 to 0.054 (0 of 12 cells outside the band). The mean null estimate

Table 20: S10. Main-effect power by biomarker correlation (rows) and design, encoding and half-life in days (columns). Dropped cells are absent. Monte Carlo standard error at most 0.022.

c_bm	Alt, mean_moderation, 0 d	Alt, mean_moderation, 3 d	Alt, mvn, 0 d	Alt, mvn, 3 d	Hybrid, mean_moderation, 0 d	Hybrid, mean_mode
0.0	0.990	0.988	0.980	0.978	0.830	0.772
0.2	0.986	0.984	0.988	0.978	0.808	0.736
0.4	0.980	0.966	0.992	0.978	0.788	0.746
0.6	0.964	0.912			0.788	0.704

Table 21: S11. Rejection rate by data-generating condition (rows) and design with assumed half-life (columns). Monte Carlo standard error at most 0.022.

DGP condition	Alt, 0.00 wk	Alt, 0.20 wk	Alt, 0.43 wk	Hybrid, 0.00 wk	Hybrid, 0.20 wk	Hybrid, 0.43 wk	RCT, 0.00 wk	RC
alt: carryover off, AR(1) high	1.000	1.000	1.000	0.974	0.972	0.966	0.504	0.5
alt: carryover off, AR(1) off	0.990	0.990	0.994	0.818	0.824	0.828	0.928	0.9
alt: carryover on, AR(1) high	0.992	0.994	0.994	0.934	0.934	0.946	0.514	0.5
alt: carryover on, AR(1) off	0.980	0.984	0.996	0.732	0.738	0.780	0.890	0.8
null: carryover off, AR(1) high	0.058	0.058	0.066	0.042	0.040	0.038	0.046	0.0
null: carryover off, AR(1) off	0.104	0.108	0.148	0.080	0.080	0.074	0.038	0.0
null: carryover on, AR(1) high	0.060	0.062	0.070	0.046	0.044	0.046	0.046	0.0
null: carryover on, AR(1) off	0.118	0.120	0.150	0.038	0.038	0.042	0.054	0.0

Table 22: S12. Type I error by configuration (rows) and design with assumed half-life (columns). Monte Carlo standard error at most 0.008.

Configuration	Alt, 0.00 wk	Alt, 0.20 wk	Hybrid, 0.00 wk	Hybrid, 0.20 wk	RCT, 0.00 wk	RCT, 0.20 wk	XO, 0.00 wk	XO, 0.20 w
1	0.092	0.095	0.054	0.054	0.049	0.049	0.162	0.162
2	0.073	0.075	0.057	0.057	0.059	0.059	0.137	0.137
3	0.075	0.080	0.070	0.069	0.049	0.049	0.120	0.120
4	0.120	0.127	0.049	0.049	0.058	0.058	0.174	0.174

ranged from Alt -0.115 to -0.029, Hybrid -0.000 to 0.018, and RCT -0.013 to 0.005. Under the alternative, power ranged from Alt 0.980 to 1.000, Hybrid 0.732 to 0.974, and RCT 0.504 to 0.928. Each cell has 500 replicates.

4.13 S12. Null calibration

We simulated a null drug effect in 4 configurations of the baseline SD, the drug-response SD and the half-life, with 2,000 replicates per cell. Pooled over configurations, Type I error was as follows. Alt 0.090 (MCSE 0.003, 16.5 MCSE from 0.05) at analysis half-life 0.00 and 0.094 (MCSE 0.003, 18.2 MCSE from 0.05) at analysis half-life 0.20. Hybrid 0.057 (MCSE 0.003, 3.0 MCSE from 0.05) at analysis half-life 0.00 and 0.057 (MCSE 0.003, 2.9 MCSE from 0.05) at analysis half-life 0.20. RCT 0.054 (MCSE 0.003, 1.5 MCSE from 0.05) at analysis half-life 0.00 and 0.054 (MCSE 0.003, 1.5 MCSE from 0.05) at analysis half-life 0.20. XO 0.148 (MCSE 0.004, 40.3 MCSE from 0.05) at analysis half-life 0.00 and 0.148 (MCSE 0.004, 40.3 MCSE from 0.05) at analysis half-life 0.20. At analysis half-life 0.2 week, Type I error by configuration was Alt 0.075 to 0.127 (mean estimate -0.067 to -0.059, κ 0.965 to 0.982), Hybrid 0.049 to 0.069 (mean estimate 0.013 to 0.019, κ 0.941 to 0.999), RCT 0.049 to 0.059 (mean estimate -0.001 to 0.001, κ 0.965 to 1.015), and XO 0.120 to 0.174 (mean estimate -0.149 to -0.139, κ 0.828 to 0.866). The MCSE of a mean estimate was at most 0.009.

S12b fitted the same null datasets with and without the serial-correlation structure. Type I error and mean estimate were 0.100 (MCSE 0.007) and -0.091 for Alt without corCAR1, 0.104 (MCSE 0.007) and -0.066 for Alt with corCAR1, 0.143 (MCSE 0.008) and -0.140 for XO without corCAR1, and 0.143 (MCSE 0.008) and -0.140 for XO with corCAR1. Each cell has 2,000 replicates.

4.14 Mean of the estimate under the null

For the blinded within-subject designs the baseline row has expectancy 0 and every follow-up visit 0.5, and the model has no term for this step. Script `18-null-shift-check.R` fitted one large null dataset per design four ways. The coefficient of D_{bc} was as follows. XO (6,000 participants): -0.109 (SE 0.018) with the fitted model, 0.017 (SE 0.018) with the fitted model plus a follow-up indicator, 0.017 (SE 0.023) with the fitted model without the baseline row, and 0.017 (SE 0.018) with visit as a factor in place of linear time. Alt (6,000 participants): -0.051 (SE 0.009) with the fitted model, 0.010 (SE 0.009) with the fitted model plus a follow-up indicator, 0.010 (SE 0.010) with the fitted model without the baseline row, and 0.011 (SE 0.009) with visit as a factor in place of linear time. Hybrid (8,000 participants): 0.014 (SE 0.011) with the fitted model, 0.022 (SE 0.011) with the fitted model plus a follow-up indicator, 0.024 (SE 0.012) with the fitted model without the baseline row, and -0.009 (SE 0.015) with visit as a factor in place of linear time.

5 Discussion

5.1 Principal findings

At the baseline effect in S1, designs ordered by power were Alt 0.983 (MCSE 0.004), Compact2 0.892 (MCSE 0.010), RCT 0.788 (MCSE 0.013), Hybrid 0.780 (MCSE 0.013), XO 0.758 (MCSE 0.014), RCT-OL 0.744 (MCSE 0.014). The baseline cell recurs in 8 sweeps with independent draws. Across them, power ranged over 0.966 to 0.990 for Alt, 0.752 to 0.824 for Hybrid, and 0.776 to 0.834 for RCT. Pooled over these cells, power was Alt 0.980 (MCSE 0.002) on 4,500 replicates, Hybrid 0.778 (MCSE 0.006) on 4,500 replicates, and RCT 0.798 (MCSE 0.006) on 4,500 replicates. The smallest N in the grid with power of at least 0.80 was Alt 16, Hybrid 60, and RCT 35.

The ANCOVA arm contrast of the parallel group behaved as intended under the null. Its mean null estimate was within 0.013 units of zero in every null cell of S11, S12 and S14. Its κ lay between 0.96 and 1.05. Pooled over the S12 configurations, its Type I error was 0.054 (MCSE 0.003), 1.5 MCSE above 0.05. One of the twelve S14 ANCOVA null cells, the open-label parallel group at $\rho = 0$, had 0.065 (MCSE 0.008), just above the band. The mixed-model analysis of the 2026-05 version had mean null estimates from -0.54 to -0.36 and Type I error from 0.17 to 0.40 in the same data. The correction removes the bias of the comparator that the referee found.

The Hybrid was slightly anticonservative. Pooled over the S12 configurations, its Type I error was 0.057 (MCSE 0.003), 2.9 MCSE above 0.05. By configuration it ranged from 0.049 to 0.069. The null cell of S1 had 0.078 (MCSE 0.008). Its mean null estimate was between 0.013 and 0.019 units, and its κ between 0.93 and 1.00. The excess of about one percentage point reported by the referee is therefore still present. By a normal approximation, model-based standard errors 3 to 7 percent too small account for most of it, and the mean shift for little.

The alternating design and the crossover were anticonservative by more. Pooled over the S12 configurations, their Type I error was 0.094 (MCSE 0.003) and 0.148 (MCSE 0.004). Their mean null estimates were about −0.06 and −0.14 units, in the direction of benefit. Removing `corCAR1` did not remove the excess (S12b). The check of Section 4.14 located the cause in the mean model. With a follow-up indicator, without the baseline row, or with visit as a factor, the large-sample null coefficient was within 0.02 units of zero for both designs. For the crossover, κ of 0.83 to 0.87 adds to the excess. Part of the power of these two designs at the alternative therefore comes from the analysis model. This study does not estimate how large that part is. The empirical SE of the alternating design at the baseline was 0.126, against 0.168 for the Hybrid. That suggests that its advantage would not vanish under a corrected model, but this was not tested.

At the baseline the Hybrid and the parallel group differed in power by 0.008 in S1, with an MCSE of about 0.013 for each. Pooled over the eight sweeps that contain the baseline cell, the parallel group had 0.020 more power than the Hybrid, with an MCSE of about 0.009 for the difference. In S7 the parallel group reached 0.80 at N = 35 and the Hybrid at N = 60, the next grid value. The grid does not locate the Hybrid's crossing point between those values.

Carryover lowered the power of the within-subject designs and not of the parallel group. As the data-generating half-life rose from 0 to 14 days, Hybrid power fell from 0.840 to 0.362 and its mean estimate moved from −0.50 to −0.27, against a fixed estimand of −0.50. The two assumed half-lives of S2, 0 and 0.2 week, changed power by at most 0.026. At 3 days in S3, the data-generating half-life in the analysis raised Hybrid power over the binary indicator by 0.010 to 0.024 across decay forms. These are paired comparisons on the same datasets. In the same sweep the decay form changed Hybrid power by up to 0.08 at a fixed analysis. S2 does not include the data-generating half-life as an analysis value, so the share of the loss above 3 days that a correct analysis half-life would recover is not estimated. In S11 the data-generating analysis half-life raised the Type I error of the alternating design without carryover from 0.108 to 0.148 (MCSE about 0.015).

Treatment heterogeneity lowered the power of the parallel group from 0.810 to 0.412 as τ rose from 0 to 1, with κ near 1. The Hybrid fell from 0.766 to 0.692. At $\tau = 1$ its κ was 0.815 and its coverage 0.888, because the model has no random slope for D_{bc}. The Hybrid's power under heterogeneity is therefore overstated. A test with this standard error would also reject a true null too often if heterogeneity were present. The DGP cannot show this, because it scales the heterogeneity by the drug-response curve, which is zero at the null. The scaling divides by the asymptote, and the asymptote differs by design at the matched estimand. The effective heterogeneity SD at the mean on-drug visit is therefore 0.38τ for the alternating design, 0.76τ for the Hybrid and 0.89τ for the parallel group. The designs are therefore not compared at the same heterogeneity in S4.

Serial correlation lowered parallel-group power from 0.922 at $\rho = 0$ to 0.336 at $\rho = 0.9$. Hybrid power rose from 0.764 at $\rho = 0$ to 0.910 at $\rho = 0.7$. The Hybrid and the alternating design were infeasible at $\rho = 0.9$. Under the mean-shift encoding the biomarker term adds participant-level variation to the drug response. Parallel-group power then fell from 0.776 to 0.592 as c_{bm} rose from 0 to 0.6 at half-life 0, and κ of the alternating design fell to 0.79 to

0.80. Under the correlation encoding, main-effect power changed by at most 0.036 between $c_{bm} = 0$ and 0.4, within two MCSE of the difference.

5.2 Relation to earlier work

The sample-size comparison of Blackston et al.[4] used a parallel group with one observation per participant. That comparator is unaffected by carryover by construction, and our parallel group has eight observations per participant, so the two are not the same comparison. Wang and Schork[5] studied one participant with 400 observations and no parallel group. They found that serial correlation lowers the power of the within-subject design and that stacking periods mitigates the loss. Our S8 and S9 vary the correlation and the period length in a series of participants, with the effect held fixed across designs. Percha et al.[6] studied the number of blocks and the sampling frequency under carryover. S6 and S9 are closest to their work, and our designs differ in that the effect is matched across designs. Chen and Chen[7] and Araujo et al.[8] concern the error structure and the participant-by-treatment interaction. S4 adds that interaction to the DGP as τ. Gärtner et al.[10] compared analysis methods under carryover. S2, S3 and S11 compare analysis half-lives under carryover and under a wrong decay form, which is a different question. Carrozzo et al.[11] compared a two-arm crossover trial, not a parallel-group trial, with a meta-analysis of N-of-1 trials.

The original DGP of Hendrickson et al. holds the biomarker-response correlation constant after first exposure. The interaction estimand is then zero at a positive half-life (`docs/36`, `docs/40`). Interaction power at short half-lives in that setting is therefore a property of the original rule. It is not a result about the main effect, and this paper does not compare with it. The comparison of the two encodings of the interaction belongs to paper 01 of this series.

The standard texts on the analysis of two-period crossover trials are Jones and Kenward[19] and Senn[20]. A crossover with one assessment per period gives two follow-up observations per participant, and the baseline row makes three. S12b tested whether fitting `corCAR1` to these data changes the Type I error, by refitting without it. It did not. The crossover had 0.143 (MCSE 0.008) with and without it, and the same estimates in every replicate.

5.3 Limitations

The study is a simulation at one baseline. Its conclusions apply to the parameter values simulated.

The effect is the mean drug response on drug, matched across designs. A trial planner may instead hold the drug and dose fixed, in which case the designs differ in exposure and the comparison here does not apply.

The analysis model of the blinded within-subject designs has linear time and no term for the change in mean from baseline to follow-up. Under the null their estimates are shifted toward benefit, and their Type I error is 0.09 for the alternating design and 0.15 for the crossover. Their power is reported as computed. It is not a valid comparison with the Hybrid and the

parallel group until the sweeps are rerun with a corrected model. The correction is a change to the engine and was not made in this version.

The `lme` Wald tests at $N = 35$ use containment degrees of freedom. In the null cells of the within-subject designs κ ranged from 0.83 to 1.00. The model has no random slope for D_{bc}, and under heterogeneity κ of the Hybrid fell to 0.82. A small-sample correction such as Kenward and Roger[21] or cluster-robust standard errors with Satterthwaite degrees of freedom may be needed. The corrections are not applied.

The large-sample value `ETbeta` is unreliable for the alternating design in some cells (Section 3.6). The heterogeneity of S4 is scaled by each design's asymptote, so a given τ is a different amount of heterogeneity in each design. S2 crosses the data-generating half-life with assumed half-lives of 0 and 0.2 week only. The data-generating value is an analysis option only at 3 days, in S3 and S11.

The parallel group has 17 or 18 participants per arm. Its ANCOVA uses the baseline score and the arm only.

The Hybrid is the only design with open-label visits. The comparators are blinded throughout, and the contrast of `RCT` with `RCT-OL` in S1 measures the effect of that choice for the parallel group only. The effect of an open-label phase on the alternating design is not measured.

S6 and S9 use the alternating design, because the Hybrid has a fixed schedule. Their implications for the Hybrid are inferences.

The carryover rule compounds across consecutive off-drug visits, so the half-life axes are nominal. A DGP in which the response does not restart from zero at re-exposure (`docs/38`) is not simulated. The study has no dropout and no missing data. The designs not evaluated are the single-participant N-of-1, the open-label-only design, multi-cycle designs with more than two periods per sequence, Latin-square designs and sequential designs[22,23]. The project whitepapers `docs/39` and `docs/40` report that more switches can lower the power of the interaction test under compound-symmetry correlation with carryover. This paper does not extend that result to the main effect, and it makes no claim of a bound for the unevaluated designs.

5.4 Open items

Three decisions are open. The first is whether paper 05 should quote half-lives in weeks, to match papers 01 to 04. The second is the baseline effect of -0.5 outcome units, chosen because a local test with 200 replicates gave the Hybrid a power below 1.000 at this value, which the earlier baseline did not. The third is the author list on the title page.

Three analyses remain to be run. The first is a rerun of the within-subject designs with a term for the change from baseline to follow-up, or with visit as a factor. It applies to every sweep that contains the alternating design or the crossover, and to the Hybrid for consistency. The second is an analysis of the Hybrid with a random slope for D_{bc} or cluster-robust standard errors, in S4 and S12. The third is the data-generating half-life as an analysis option at every half-life of S2.

6 Conclusions

With the designs matched on the mean drug response on drug and N = 35, the Hybrid had 0.020 less power than a parallel group analyzed by ANCOVA, with an MCSE of 0.009 for the difference. This is pooled over the eight sweeps that contain the baseline cell. The parallel group's test held its size. The Hybrid's test exceeded nominal by about one percentage point.

The two designs responded to different factors. Carryover lowered Hybrid power from 0.840 to 0.362 between half-lives of 0 and 14 days and left the parallel group unchanged. Treatment heterogeneity and serial correlation lowered parallel-group power. The Hybrid's model-based SE understated its variability under heterogeneity.

The alternating design had the highest power. The crossover had the highest Type I error. Both analyses were shifted under the null by an unmodeled change from baseline to follow-up. The ordering of these two designs against the Hybrid and the parallel group is therefore not established by this run.

Four methodological points hold regardless of the parameter values. A comparison of N-of-1 and parallel designs needs a common estimand. A comparator needs an analysis whose null behavior has been checked, and the same check applies to every within-subject design. Cells with a covariance matrix that is not positive definite need to be dropped or reported, not repaired. Heterogeneity should be defined on the scale of the estimand if designs are to be compared at the same heterogeneity. The study is exploratory, and its conclusions apply to the parameter values simulated.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The drivers, the engine and the stored artifacts are in the pmsimstats-ng project repository, under `analysis/scripts/nof1-design-sensitivity/`, `implementations/nof1power/` and `analysis/data/derived_data/sensitivity/`. A public repository address has not been assigned. It will be added at submission.

10 Reproducibility

The manuscript is rendered with `bash tools/render.sh analysis/report/05-nof1-design-sensitivity/r`
The sweeps are run with the commands in `README.md` of this directory after `devtools::install_local('implem`
Each driver sets its seed with `set.seed(42 + id)`, checks that the parameter values stored in the artifact equal the grid, and saves an artifact with the engine version, the repository revision and the replicate count. If a check fails the artifact is saved under a name ending in `-UNCHECKED.rds`, and this document ignores it. No such file exists for the stored run. The check of Section 4.14 is `18-null-shift-check.R`, which fits one large null dataset per design and takes about two minutes. Rendering this document does not run any simulation.

The principal packages are `nlme` for the analysis, `corpcor` for the positive-definiteness check, `furrr` and `future` for parallel execution, and `dplyr` and `tidyr` for summaries. The package set is recorded in `renv.lock`. The R version used for this render is R version 4.6.1 (2026-06-24).

11 Glossary

- **Alt.** The alternating design: two paths that switch drug state at every visit.
- **Analysis half-life.** The half-life $t^a_{1/2}$ used to build the exposure indicator D_{bc} in the analysis. It need not equal the data-generating half-life.
- **ANCOVA arm contrast.** The analysis of the parallel group: follow-up mean regressed on arm and baseline score.
- **`br_max`, `br_sd`, `bl_sd`, `pb_max`, `tv_max`.** Stored values of the drug-response asymptote and SD, the baseline-score SD, and the placebo-response and natural-history asymptotes.
- **`c.bm`, c_{bm}.** The correlation between the biomarker and the drug response at on-drug visits in the correlation encoding.
- **Compact2.** The two-path schedule of eight weekly visits used in the 2026-05 version. It is not the published Hybrid.
- **`delta_on`, Δ_{on}.** The realized estimand: the mean drug response over the on-drug follow-up visits of a design.
- **`Dbc`, D_{bc}.** The exposure indicator of the analysis: 1 on drug, decaying off drug after exposure, 0 before exposure.
- **`ETbeta`.** The large-sample value of the estimated coefficient: the coefficient of a fit to all simulated participants of a cell.
- **Feasibility.** A cell is feasible when its covariance matrix is positive definite. The engine drops a cell that is not.
- **Hybrid.** The design of Hendrickson et al.: open-label titration, blinded discontinuation, then a brief blinded crossover. Four paths, eight visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **κ.** The mean model-based standard error divided by the empirical SD of the estimates. A value of 1 means calibrated.

- **`legacy`.** The mixed-model analysis of the parallel group used in the 2026-05 version.
- **`mean_moderation`.** The mean-shift encoding of the biomarker-by-drug interaction.
- **`mvn`.** The correlation encoding of the interaction: the interaction is a correlation between the biomarker and the drug response that is present on drug and absent off drug.
- **MCSE.** Monte Carlo standard error.
- **`pd_repaired`.** Stored flag, `TRUE` if a covariance matrix was repaired. It is `FALSE` for every stored replicate.
- **RCT, RCT-OL.** The parallel group with blinded expectancy 0.5, and with open-label expectancy 1.
- τ**.** The SD of participant-level heterogeneity of the drug effect.
- **XO.** The two-period crossover with one assessment per period.

12 Abbreviations

ADEMP, aims, data-generating mechanisms, estimands, methods and performance measures; ANCOVA, analysis of covariance; AR(1), first-order autoregressive; BR, drug-response component; DGP, data-generating process; MCSE, Monte Carlo standard error; PB, placebo-response component; RCT, randomized controlled trial; SD, standard deviation; SE, standard error; TV, natural-history component.

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